<u>OSSP-1</u>

<u>Question:</u>

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Solution:

```
GNU nano 8.7.1                                        practical1.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    char command[100];

    printf("Enter Linux command: ");
    scanf("%s", command);

    pid = fork();

    if(pid < 0)
    {
        printf("Fork failed\n");
        return 1;
    }
    else if(pid == 0)
    {
        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());

        execlp(command, command, NULL);

        printf("Command execution failed\n");
    }
    else
    {
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());

        wait(NULL);
```

Output:

```
ubuntu@ubuntu:~$ nano practical1.c
ubuntu@ubuntu:~$ gcc practical1.c -o practical1
ubuntu@ubuntu:~$ ./practical1
Enter Linux command: ls

Parent Process
Parent PID: 7084

Child Process
Child PID: 7086
Desktop    Downloads  Pictures  Templates  practical1    snap
Documents  Music      Public    Videos     practical1.c
Child process completed
```

Linux terminal commands:

```
ubuntu@ubuntu:~$ uname -a
Linux ubuntu 7.0.0-14-generic #14-Ubuntu SMP PREEMPT_DYNAMIC Mon Apr 13 10:52:31 UTC 2026 aarch64 GNU/Linux
```

```
ubuntu@ubuntu:~$ lscpu
Architecture:             aarch64
  CPU op-mode(s):         64-bit
  Byte Order:             Little Endian
CPU(s):                   4
  On-line CPU(s) list:    0-3
Vendor ID:                Apple
  Model name:             -
    Model:                0
    Thread(s) per core:   1
    Core(s) per socket:   4
    Socket(s):            1
    Stepping:             0x0
    BogoMIPS:             48.00
    Flags:                fp asimd evtstrm aes pmull sha1 sha2 crc32 atomics
                          fphp asimdhp cpuid asimdrdm jscvt fcma lrcpc dcpop
                          sha3 asimddp sha512 asimdfhm dit uscat ilrcpc flagm
                           sb paca pacg dcpodp flagm2 frint bf16 bti
NUMA:
  NUMA node(s):           1
  NUMA node0 CPU(s):      0-3
Vulnerabilities:
  Gather data sampling:   Not affected
  Ghostwrite:             Not affected
  Indirect target selection: Not affected
  Itlb multihit:          Not affected
  L1tf:                   Not affected
  Mds:                    Not affected
  Meltdown:               Not affected
  Mmio stale data:        Not affected
  Old microcode:          Not affected
  Reg file data sampling: Not affected
  Retbleed:               Not affected
  Spec rstack overflow:   Not affected
  Spec store bypass:      Vulnerable
  Spectre v1:             Mitigation; __user pointer sanitization
  Spectre v2:             Mitigation; CSV2, but not BHB
  Srbds:                  Not affected
  Tsa:                    Not affected
  Tsx async abort:        Not affected
```

```
ubuntu@ubuntu:~$ lsblk
NAME    MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0     7:0    0   2.9G  1 loop /rofs
loop1     7:1    0 492.6M  1 loop
loop2     7:2    0 138.1M  1 loop
loop3     7:3    0  61.9M  1 loop /snap/core24/1588
loop4     7:4    0  19.5M  1 loop /snap/desktop-security-center/149
loop5     7:5    0     4K  1 loop /snap/bare/5
loop6     7:6    0 265.3M  1 loop /snap/firefox/8105
loop7     7:7    0 552.9M  1 loop /snap/gnome-46-2404/154
loop8     7:8    0  91.7M  1 loop /snap/gtk-common-themes/1535
loop9     7:9    0  14.9M  1 loop /snap/snap-store/1368
loop10    7:10   0 174.6M  1 loop /snap/mesa-2404/1166
loop11    7:11   0  17.5M  1 loop /snap/prompting-client/203
loop12    7:12   0  42.6M  1 loop /snap/snapd/26869
loop13    7:13   0   221M  1 loop /snap/thunderbird/1045
loop14    7:14   0 109.4M  1 loop /snap/ubuntu-desktop-bootstrap/590
loop15    7:15   0   560K  1 loop /snap/snapd-desktop-integration/363
sr0      11:0    1   3.9G  0 rom  /cdrom
vda     253:0    0    12G  0 disk
```

```
ubuntu@ubuntu:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 05:06 ?        00:00:04 /sbin/init --- splash
root           2       0  0 05:06 ?        00:00:00 [kthreadd]
root           3       2  0 05:06 ?        00:00:00 [pool_workqueue_release]
root           4       2  0 05:06 ?        00:00:00 [kworker/R-rcu_gp]
root           5       2  0 05:06 ?        00:00:00 [kworker/R-sync_wq]
root           6       2  0 05:06 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root           7       2  0 05:06 ?        00:00:00 [kworker/R-slub_flushwq]
root           8       2  0 05:06 ?        00:00:00 [kworker/R-netns]
root          10       2  0 05:06 ?        00:00:00 [kworker/0:0H-kblockd]
root          13       2  0 05:06 ?        00:00:00 [kworker/R-mm_percpu_wq]
root          14       2  0 05:06 ?        00:00:00 [ksoftirqd/0]
root          15       2  0 05:06 ?        00:00:01 [rcu_preempt]
root          16       2  0 05:06 ?        00:00:00 [rcu_exp_par_gp_kthread_worker/0]
root          17       2  0 05:06 ?        00:00:00 [rcu_exp_gp_kthread_worker]
root          18       2  0 05:06 ?        00:00:00 [migration/0]
root          19       2  0 05:06 ?        00:00:00 [idle_inject/0]
root          20       2  0 05:06 ?        00:00:00 [cpuhp/0]
root          21       2  0 05:06 ?        00:00:00 [cpuhp/1]
root          22       2  0 05:06 ?        00:00:00 [idle_inject/1]
root          23       2  0 05:06 ?        00:00:00 [migration/1]
root          24       2  0 05:06 ?        00:00:00 [ksoftirqd/1]
root          26       2  0 05:06 ?        00:00:00 [kworker/1:0H]
root          27       2  0 05:06 ?        00:00:00 [cpuhp/2]
root          28       2  0 05:06 ?        00:00:00 [idle_inject/2]
root          29       2  0 05:06 ?        00:00:00 [migration/2]
root          30       2  0 05:06 ?        00:00:00 [ksoftirqd/2]
root          32       2  0 05:06 ?        00:00:00 [kworker/2:0H-kblockd]
root          33       2  0 05:06 ?        00:00:00 [cpuhp/3]
root          34       2  0 05:06 ?        00:00:00 [idle_inject/3]
root          35       2  0 05:06 ?        00:00:00 [migration/3]
root          36       2  0 05:06 ?        00:00:00 [ksoftirqd/3]
root          38       2  0 05:06 ?        00:00:00 [kworker/3:0H-kblockd]
root          39       2  0 05:06 ?        00:00:00 [kdevtmpfs]
root          40       2  0 05:06 ?        00:00:00 [kworker/R-inet_frag_wq]
root          41       2  0 05:06 ?        00:00:00 [rcu_tasks_kthread]
root          42       2  0 05:06 ?        00:00:00 [rcu_tasks_rude_kthread]
root          43       2  0 05:06 ?        00:00:00 [kauditd]
root          44       2  0 05:06 ?        00:00:00 [khungtaskd]
```

```
ubuntu@ubuntu:~$ top

top - 07:52:41 up  2:46,  1 user,  load average: 0.15, 0.10, 0.04
Tasks: 229 total,   1 running, 228 sleeping,   0 stopped,   0 zombie
%Cpu(s):  1.2 us,  0.5 sy,  0.0 ni, 98.3 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3892.8 total,     88.5 free,   2741.7 used,   1515.9 buff/cache
MiB Swap:      0.0 total,      0.0 free,      0.0 used.   1151.1 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   3356 ubuntu    20   0 5173088 486056 165344 S   2.3  12.2   0:57.22 gnome-shell
   5708 ubuntu    20   0 2929028 490636 136412 S   2.3  12.3   0:24.95 ptyxis
     48 root      20   0       0      0      0 S   0.7   0.0   0:01.72 kcompactd0
   4103 ubuntu    20   0  404472  20068   6744 S   0.3   0.5   0:01.34 gvfs-afc-volume
   7295 ubuntu    20   0   22008   5156   3088 R   0.3   0.1   0:00.02 top
      1 root      20   0   26656  17064  10552 S   0.0   0.4   0:04.07 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.01 kthreadd
      3 root      20   0       0      0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
      7 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-netns
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-kblockd
     13 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
     14 root      20   0       0      0      0 S   0.0   0.0   0:00.25 ksoftirqd/0
     15 root      20   0       0      0      0 I   0.0   0.0   0:01.01 rcu_preempt
     16 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
     17 root      20   0       0      0      0 S   0.0   0.0   0:00.04 rcu_exp_gp_kthread_worker
     18 root      rt   0       0      0      0 S   0.0   0.0   0:00.15 migration/0
     19 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/0
     20 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     21 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
     22 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/1
     23 root      rt   0       0      0      0 S   0.0   0.0   0:00.15 migration/1
     24 root      20   0       0      0      0 S   0.0   0.0   0:00.13 ksoftirqd/1
     26 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworker/1:0H
     27 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/2
     28 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_inject/2
     29 root      rt   0       0      0      0 S   0.0   0.0   0:00.14 migration/2
     30 root      20   0       0      0      0 S   0.0   0.0   0:00.16 ksoftirqd/2
```